

CITATIONFLOW ANALYTICS

Research Citation Network Analysis and Influence Ranking System

Team no : 24

G Jahnavi-2520030483

Ananya roy-2520030471

ABSTRACT

- CitationFlow Analytics analyzes relationships between research papers.
 - Each research paper is represented as a node.
 - A citation between two papers is represented as a directed edge.
 - The system builds a citation network to study the flow of knowledge.
 - It identifies influential research papers based on citation relationships.
 - It also visualizes citation paths and generates analytical reports.
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PROBLEM STATEMENT

- There are a large number of research papers published every year.
- Understanding the relationships between papers manually is difficult.
- It is hard to identify influential research papers in a large citation network.
- Tracking how one research paper influences other papers can be complex.
- A system is needed to efficiently analyze citation relationships and knowledge flow.

PROPOSED FUNCTIONALITIES

The system will provide:

- Add and manage research papers
- Create citation relationships between papers
- Build a directed citation graph
- Identify influential research papers
- Analyze citation paths
- Calculate citation flow and paper rankings
- Visualize the citation network
- Generate citation analysis reports

Tools and Technologies

- Java
- Directed Graph
- Adjacency List
- BFS
- DFS
- Edmonds-Karp (Maximum Flow)
- Priority Queue / Sorting
- Java Collections Framework
- HTML
- CSS
- JavaScript
- Java HttpServer

Expected Outcome

- Create a network of research papers and citations
- Represent citation relationships using a directed graph.
- Identify important and influential research papers.
- Show how knowledge flows through the citation network.
- Display citation paths and relationships visually.
- Generate rankings and analytical reports.
- Efficiently handle multiple research papers and citation relationships.

THANK YOU
